

90 Hypotheses, Zero Edges

I built a pre-registered research pipeline to find tradeable edges for a £1,500 retail account.

It correctly found nothing. This is the audit trail.

Nothing described here traded real money.

All results are paper or backtest. This is a research log, not advice.

It correctly found nothing

The claim

Over several months I built an automated research pipeline to find tradeable edges accessible to a retail investor with $\sim$ £1,500 in a UK ISA. It tested roughly 100 ideas across prediction markets, corporate actions, index flows, insider filings and ETF cost structures.

It found no edge that pays income at that scale. That is the result. This book is about why I believe it, which is the only part worth reading.

Why a negative result here is worth anything

Most retail backtests find edges. Almost all of those edges are artifacts. The interesting question is not "did you find something" but "would you have known if you hadn't."

Five mechanisms made the answer trustworthy.

1. Pre-registration with a cited direction. Hypotheses were registered with their predicted sign before data was pulled, hash-chained so the registration cannot be revised after the fact. A sentiment signal registered as "attention predicts returns, direction per Barber & Odean 2008" came back at $t = 0.04$. Without pre-registration the honest move — reporting nothing — is also the least appealing one.

2. Multiple-testing correction across the true m . With $m = 38$ hypotheses and FDR $q = 0.05$, the Benjamini-Hochberg raw- p bar is **0.00685**, not 0.05. Testing 155 strategy variants at $p < 0.05$, as one popular method does, yields ~ 7.8 false positives by chance alone — so "5 survivors from 155" is *below* what pure noise produces.

3. Overlap-aware inference. With 63–252 day holding periods entered continuously, dozens of positions are open simultaneously and driven by the same market. Counting them as independent overstates the evidence by roughly the overlap factor — measured at **17–27x** on the real sets. Collapsing to calendar-time monthly returns (Fama 1998) and bootstrapping that series moved one hypothesis from $p = 0.049$ to **$p = 0.87$** .

4. Factor attribution before celebrating a return. A buyback-announcement strategy showed 14.5% raw. Against a six-factor model: **alpha 4.6%, beta 1.0047, R^2 0.96**. The raw number was almost entirely market exposure. Separately, the three largest "alphas" in the batch had **$R^2 = 0.00$** — failed regressions, not discoveries. Any ranking by alpha without checking R^2 puts the most broken results on top.

5. Partial identification instead of assuming the missing data away. Where $\sim 46\%$ of events were unobservable, a Manski bound gives what the result could be under any assumption about the missing half, rather than silently conditioning on survivors.

The machine that stops you lying

The pipeline's primary job was not running backtests. It was **making it impossible to fool myself while running hundreds of them**. At a 5% significance level, 100 uncorrected tests yield ~ 5 false discoveries by pure chance; 500 yield ~ 25 — and they look exactly like real edges: clean equity curves, good p-values, plausible stories. Every design decision below serves that problem.

The four non-negotiables

1. Pre-registration before testing. The claim, direction, universe, holding period, success criterion and economic rationale are recorded — timestamped, **before any data is touched**. The runner refuses ids with no registration record. The log is append-only and hash-chained: editing, deleting or reordering any row breaks chain verification, which every command runs before trusting anything. A failed hypothesis stays in the log forever and counts against the FDR correction forever. This is the difference between "I predicted this" and "I found this."

2. FDR correction, automatic and unavoidable. Benjamini-Hochberg is recomputed across **every hypothesis ever tested in the project** — latest result per id, all sessions — on every read. The rendered report never shows a raw p-value without its adjusted counterpart on the same line, enforced by test. Bonferroni was rejected as too conservative at these counts; BH controls the false discovery rate, which is the quantity that matters when the point is choosing what to trust.

3. Sealed holdout, one shot per hypothesis. History splits at a **fixed calendar date**, not a percentage, so a longer download cannot silently move data between pools. A hypothesis touches holdout **exactly once**, only after passing development. The attempt is burned in the ledger **before** the test runs, so a crash mid-test still consumes it — "it crashed, run it again" is a re-test loophole and it is closed. Fail holdout and the hypothesis is dead: not re-tunable, not re-testable.

4. No parameter sweeping, enforced rather than documented. Parameters are scalars fixed at registration; a list, range or dict parameter value is refused as the registration-time signature of a sweep. A test fails if sweep vocabulary appears anywhere in the harness source. Three variants of an idea are three registrations counting three times against FDR — which is the correct accounting, not an inconvenience.

What the machine still cannot do

Stated here rather than buried, because a book about self-deception that hides its own caveats is not worth reading.

- **BH assumes independent tests, and these are not independent.** Calendar hypotheses share the same index; EDGAR event studies share a filer universe; event windows overlap in time. The effective number of independent tests is lower than nominal m , which makes the displayed bar *more* conservative than exact. It is stated wherever the threshold renders, so a survivor is never declared off a bar read as precise.
- **Nearly every equity universe here is survivorship-biased.** Upper bounds only.
- **Free fundamentals are not point-in-time**, which is why no fundamental-signal evaluator was ever built.
- **Factor data lags ~2 months**, so attribution over recent windows silently covers fewer days.

Chapter 3

The phantom £74,000

The scanner had a historical sample: 81 filings, January 2023 to December 2025. It reported fourteen positive candidates worth \$79,978.

I validated it against itself and found that **98.6% of the apparent profit was artifacts**.

	count	total est. profit
raw "positive estimate" candidates	14	\$79,978
— of which ARTIFACTS	3	\$78,865
real candidates	11	\$1,113

Two mistakes produced the whole of it.

COKE, \$74,417 — a stock split. The tender price of \$850 was compared against a \$94.53 market price. The \$94.53 was split-adjusted; the 10-for-1 split happened in 2025, after the tender. The comparison comes to a spread of roughly 800%, which is not a number any real tender produces, and the pipeline recorded it as profit.

IMO, \$4,448 across two deals — a currency mismatch. A Canadian-dollar tender priced against the USD listing.

Both are the same error in different clothes: **a price compared against a price that was never the same thing**. Neither is subtle once seen. Neither was visible in aggregate, because the aggregate is a single large number and large numbers are what you are hoping for.

After removing them, the honest figures:

universe	deals/yr	\$/yr	£/month
NYSE/Nasdaq listed (ISA-eligible)	2.3	\$253	£17
including OTC/pink (needs GIA)	3.7	\$371	£24

And structurally, this cannot be scaled: **the edge is the 99-share odd-lot cap**. Taking more destroys it. I checked whether deal flow was being missed elsewhere — SC TO-T carries only three odd-lot filings in three years, SC 13E4 zero, SC 14D9 one. Form coverage was already complete. The only remaining lever is multiple accounts, which tender terms typically prohibit and which I am not going to recommend.

The scanner was not broken. It was working perfectly, on two numbers that did not mean what the code believed they meant.

Chapter 4

Twenty-one defects, none caught by the tests

I kept a defect catalogue. Not a bug tracker — a structured record, one row per defect, with five fields: the **claim** I had believed, the **effect** of it being wrong, the **scope** the check actually excluded, what **caught** it, and whether it could be turned into a mechanised check afterwards.

Twenty-one entries over five weeks. The interesting column is `caught_by`, and the finding is what is absent from it.

Not one defect was caught by the test suite.

Not one. They were caught by hand-reading five real filings, by spot-checking ten company names known to be alive, by opening a query form and getting a subscription error, by noticing one ticker flip from a hit to a miss between runs, by asking why a risk-free strategy had six stop-loss exits. Every single one was caught by a human being asking a question that the code had not been asked.

The suite was worse than silent

Defect D03: the nightly digest reported "**420 tests OK**". Full discovery was 833 tests with **17 failures**. The green number was measured on a six-file subset — and the subset excluded almost everything that session had changed.

So the suite was not merely failing to catch defects. It was reporting green over red, repeatedly, across a session, on precisely the files under active modification.

The shape they all share

Read the `scope_excluded` field down the catalogue and the same error appears in twenty different costumes. **The check was real. The check ran. The check did not cover the case.**

- **D04** — an amendment filter keyed on `root_forms`, a field that *normalises* amendments, so no value in it ever ended in `/A`. The filter read correctly and dropped literally nothing. Fixed, it drops 5 of 34 live hits.
- **D09** — a plausibility floor at ratio 0.45 guarding against broken inputs. The upstream scanner admits nothing below 0.50, so the guarded region was unreachable. The entire admissible band was being upsized, including the 0.575 that produced a real -\$174.05 loss.
- **D15** — a threshold requiring fee-adjusted bids to sum above 1.02. Observed maximum across **94,754 rows** was 0.9959. The whole mint-and-sell path had never been triggerable on real data.
- **D05** — "a Form 25 means the company exited." A Form 25 delists a *security*, not an issuer. 52.5% false positives, including AAPL, MSFT, JNJ, WMT, V and MA labelled as delisted.
- **D11** — fuzzy name matching resolved "British Airways" to BATS.L, which is British American Tobacco. It would have attributed one firm's returns to another firm's index event, undetectably, downstream of everything.

None of these is a coding error. Every one is a **scope** error, and scope errors are invisible to tests by construction: the test asserts the check behaves correctly on the inputs you imagined, and the defect lives entirely in the inputs you did not.

The one that took the profits with it

D20 is the worst entry in the catalogue.

The Polymarket market-maker was the only profitable strategy in the project: **+\$49.69 across 28 spread captures**. I checked the claimed fills against Polymarket's public trade tape — free to do, no real orders, and it had not occurred to me to do it for months.

60% of the fills had no supporting volume at all. 68% of claimed shares were unsupported. One market had 339 trades in *eight months* while the model was declaring fills within minutes. The fills were modelled off `last_trade_price`, which does not move when nothing trades — so the model saw a stable price and inferred a liquid book, when what it was actually observing was silence.

The only strategy that made money was substantially an artifact of the thing pretending it had made money.

Three defects in my own auditing

The catalogue includes three entries whose where field names my own verification work. I want them in the book because a catalogue with no self-referential rows is not a catalogue, it is a highlight reel.

- **D16** — the first run of the guard-reachability audit used raw sums across all baskets, when the strategy gates on fee-adjusted sums and trades at most six legs. Correctly conditioned: 189 rows, not 5,602. The audit tool's first output was itself a defect.
- **D18** — I reported that a strategy "did not trade the two underpriced baskets — UNRESOLVED" without reading the strategy's own log, which contains the fill. I handed the operator a false open question.
- **D21** — investigating D20, I found that relaxing the fill window to +/-24h dropped the impossible-fill rate from 63% to 27.7%, and briefly concluded the strict test had been too harsh. But +/-24h counts volume from long after the order was gone. The true resting window is derivable from the audit log itself and its **median is 0.4 hours** — shorter than my original guess, not longer.

D21 is the one I think about. **A relaxation that appears to rescue a finding deserves exactly the same scrutiny as the finding, and it did not get it until the third pass.** Scepticism is easy to sustain against a result you want to be false and very hard to sustain against a rescue you want to be true.

What became mechanical

Eight of the twenty-one could be turned into a standing check. They collapse into two questions, and neither is about code:

****guard_is_reachable**** — can any real input actually reach this branch? Run against logged production values, not imagined ones.

****coverage_by_outcome**** — does my data cover outcomes evenly? If coverage differs by outcome, the headline number is not interpretable. This is chapter 8's screen, arrived at from the opposite direction.

The remaining thirteen have no mechanical form. They needed someone to look at the actual thing.

A green test suite is evidence that the code does what you thought it did. What you thought is the thing under examination.

The clause that was invisible twice

A scanner that finds nothing looks exactly like a market with nothing in it.

That is the sentence I wish someone had put in front of me a year earlier. Every other failure mode in this project announced itself. A crash announces itself. A wrong number announces itself eventually, usually by being too good. But a filter that silently drops the one deal worth having produces a clean, quiet, plausible run, every morning, indefinitely. There is no alert for absence.

This chapter is about one sentence in one filing that my scanner could not see, for two unrelated reasons, either of which would have been sufficient on its own.

The sentence

Cannae Holdings filed an SC TO-I on 1 March 2024. Buried in exhibit d752033dex99a1vi.htm is the clause the entire strategy exists to find:

first, the Company will purchase all odd lots of less than 100 Shares

That is textbook odd-lot priority. Holders of fewer than 100 shares get bought out ahead of proration — which is the whole edge, and the reason the edge cannot be scaled or competed away. If a scanner built to find odd-lot priority cannot find *that*, it cannot find anything.

Mine returned `grants_genuine_priority: False`.

Defect one: a single word

The pattern required the phrase `first, the Company will purchase odd lots`.

Cannae wrote "purchase **all** odd lots".

One word, sitting between purchase and odd lots, and the match failed. There was no quantifier in the pattern because the two filings I had built it against did not happen to use one.

The fix makes quantifiers optional and broadens the actor, because filings say "we", "the Company", "the Purchaser" and "Parent" interchangeably:

```
first\s*,?\s*(?:we|the\s+[\w\s]{0,24}?)\s*will\s+purchase\s+
(?:all\s+(?:of\s+the\s+)?)?[0o]dd\s+[Ll]ots
```

Defect two: a tag

Fixing the word was not enough, because the raw document does not read the way I quoted it above. It reads:


```
<I>first</I>;, the Company will purchase all odd lots of less than 100
```

My patterns describe **prose**. The input was **HTML**. A tag can fall between any two words in a sentence, and here one falls between "first" and the comma — so even the corrected pattern matches nothing.

The pipeline had a function for this, `normalize_entities`, which collapses ` ` and friends into ordinary spaces. It does not strip tags. And `fetch_filing_text`, which produced the text being matched, never called it. So the scanner spent months running prose patterns against markup and reporting, accurately and uselessly, that it had found nothing.

The fix is one line, applied before any matching:

```
text = re.sub(r"<[^>]+\>", " ", text)
```

Note what these two defects have in common, which is nothing. One is a domain-language error: I did not know how tender offers phrase the grant. The other is a data-handling error: I did not know what I was matching against. They were introduced at different times for different reasons. **Repairing either one alone would have changed nothing**, and would have left me confident I had addressed the problem.

What it cost

CNNE was worth **\$176.32** on a 99-share lot, against **\$621.56** earned across every tradeable deal in three years.

Twenty-eight per cent of the strategy, invisible. Not lost to a bad decision, a market move, or a risk I had accepted — lost to a missing word and an italic tag, in a filing I had successfully downloaded, parsed and discarded.

And I want to be precise about how it was found, because it was not found by inspection. I did not notice it. Nobody reads a passing test suite looking for the deals it silently declined. It surfaced only when I stopped testing against fixtures and ran every field against real SEC documents — a slow, tedious exercise with no deliverable at the end, which produced seven genuine defects in a single morning.

The turn

Here is the part that makes this chapter worth its place in the book.

In that same audit, in the same table of defects, sat an unrelated finding: the price parser was reading Dutch auction ranges backwards. Chapter 6 covers the mechanism. What matters here is the consequence.

Cannae was a Dutch auction. Range \$20.75 to \$23.75. Market price \$21.96.

Scored against \$23.75, the deal I had just spent two fixes recovering is worth **+\$176**. Scored against \$20.75, which is what the method always said it used, the same deal is worth **-\$120**.

The prize behind the bug was a loss. And the \$176.32 that made both CNNE fixes feel urgent was itself computed by a third defect that the same morning's audit was in the middle of finding.

What this is actually about

Three things, and only the third is about tender offers.

Dropping filters are the dangerous class. A bug that invents a deal gets caught the moment you read the filing. A bug that discards one is indistinguishable from an honest empty market, and it is *self-concealing*: the quieter your scanner gets, the more reasonable it looks. Every filter in this pipeline that can reject a row now has a test asserting a known-good row survives it. That is a cheap discipline and I adopted it two bugs too late.

Fixtures verify what you already understood. Both defects here are failures of imagination, and a fixture is a snapshot of your imagination on the day you wrote it. My tests passed throughout. They were testing that the scanner still did what I had wrongly assumed it should. Only contact with unfamiliar real documents moves that boundary.

No figure here is final, including the ones in this chapter. The £15/month is the third number to occupy that slot, and each predecessor died to a defect found by looking harder, not by looking again. I have no reason to believe the sequence has stopped.

Chapter 6

The inverted range

The defect fixed alongside the CNNE clause was not hypothetical, and it cut the headline figure by more than a third.

Filings state a Dutch auction range in either order, and the high-first phrasing is the common one:

not greater than \$23.75 nor less than \$20.75 per Share

My parser took the first number it found as the low. So every Dutch auction had been scored against the **top** of its range, while the methodology stated, in writing, that it used the conservative low end.

Re-scoring the historical sample against the true low turns four of five previously-positive Dutch auctions negative:

date	tick	old low	true low	mkt	old P&L	true P&L
2023-03-06	CORT	22.00	19.25	21.20	+79	-193
2023-06-08	DBI	8.00	7.00	8.65	-65	-163
2023-07-27	ILLMF	2.65	2.53	1.87	+77	+65
2023-11-13	OSPN	11.00	9.50	9.97	+101	-47
2023-12-01	OSPN	11.00	9.50	10.24	+75	-73
2024-03-01	CNNE	23.75	20.75	21.96	+176	-120
ex-artifact positive total, old parser				\$1,113	->	£24/month
ex-artifact positive total, TRUE low				\$ 673	->	£15/month

Six of the nine Dutch auctions in the sample were inverted.

Two things are worth saying plainly about this.

The £24 was never real. It was the same deals priced optimistically. £15/month is the first figure in this project computed the way the methodology always said it was computed. This is the fix working, not a new problem.

The differences are not marginal. CORT moved \$272 per 99-share lot and CNNE \$296. A deal whose market price sits *inside* the range flips sign entirely — and a Dutch auction is precisely the mechanism that produces that population. The error was not adding noise to the estimate. It was selecting for the deals most likely to be wrong, in the direction most likely to be flattering.

Chapter 7

Counting the same trade twenty-seven times

With holding periods of 63 to 252 days, entered continuously, dozens of positions are open simultaneously and driven by the same market. Treating each as an independent observation is not a small approximation. It is counting one piece of evidence many times.

The inflation factor is measurable: **26.5x** for one hypothesis, 22.0x for another, 17.3x for a third. Roughly speaking, 5,407 trades were carrying about as much independent information as 204 months.

The correction is standard. Collapse overlapping trades to one observation per calendar month (Fama 1998, Mitchell–Stafford), bootstrap that series with a seeded stationary bootstrap (Politis–Romano), and take $\max(p_{\text{stored}}, p_{\text{calendar}})$ — never the friendlier of the two. The method was fixed before these results existed. No threshold was tuned, no variant searched, no holdout consumed.

id	p stored	p calendar	p final	trades	months	overlap	moves?
H0005	0.00060	0.00005	0.00060	24	168	0.1	no
H0007	0.01455	0.00945	0.01455	100	100	1.0	no
H0022	0.00005	0.00400	0.00400	1,392	144	9.7	YES
H0023	0.00005	0.00105	0.00105	1,403	144	9.7	YES
H0024	0.04920	0.87236	0.87236	147	187	0.8	YES
H0084	0.00685	0.08635	0.08635	762	237	3.2	YES
H0086	0.00005	0.00525	0.00525	5,407	204	26.5	YES
H0087	0.00005	0.00295	0.00295	1,669	76	22.0	YES
H0090	0.00005	0.00005	0.00005	3,538	204	17.3	no

It had never actually run

Before the results, the embarrassing part. The pipeline had recorded, five days earlier, that overlap-aware inference was implemented and that a particular file "carries the retrospective view."

That file did not exist, and no hypothesis had been re-run since. The correction had been documented, described in a phase note, and applied to precisely nothing. It sat in the repository as a completed feature for the better part of a week.

The correction you have documented is not the correction you have run, and the gap between them is invisible unless you go looking for the output.

What the correction actually did

Here I have to resist my own headline. The dramatic number in that table is H0024, moving from $p = 0.049$ to $p = 0.87$ — a hypothesis that looked marginally significant and turned out to be nothing. It is the number I quoted in the summary, and it is true.

It also changed no verdict whatsoever. The Benjamini–Hochberg bar was 0.00685; H0024 at 0.049 had never cleared it. The most striking p-value movement in the entire exercise killed something that was already dead.

Exactly one verdict changed: H0084, from 0.00685 to 0.08635 — it had been sitting precisely at the bar. And its recorded alpha was already negative (-0.032 , $t = -0.56$, $R^2 0.39$), so the correction removed a hypothesis with no credible effect anyway. It removed it on inference grounds rather than waiting for someone to notice, which is the point of having the machinery, but let us be clear about the magnitude: **six of seven survivors still cleared.**

The strongest single result in the table is the one that did not move. H0090 sits at 0.00005 before and after, despite a **17.3x overlap**. Its significance is not an artifact of counting overlapping trades as independent, and now that is demonstrated rather than hoped.

I had already overstated the damage

Before running the project's own method, I used a crude proxy: clustering at the **year**. That reported only H0090 clearly surviving, with six hypotheses falling to $t \sim 1.4\text{--}2.0$. I believed it and wrote it down.

It was wrong, and it was wrong by choosing the wrong grain:

year clustering	9 – 31 observations
monthly collapse	76 – 237 observations

Annual buckets discard an order of magnitude more information than the correct calendar-time series. They inflate the standard error and manufacture a pessimism the data does not support.

The project's working rule says an aggregate must be checked at a finer grain before it is trusted. **The mirror case is equally real: a grain that is too coarse deflates just as reliably as one that is too fine inflates.**

And note the direction. Every other error in this book ran the *flattering* way, which is why I had learned to distrust good news. This one ran the unflattering way — and that is precisely why it went unchecked. It felt like the rigorous choice. Pessimism about your own results is not the same thing as rigour, and it is much harder to notice when it is unearned.

Limits

- The monthly collapse is an approximation. Trades carry only entry, exit and a net return, so each return is spread evenly across the months it spans. Exact for one-month holds; it understates within-trade path variance for longer ones. It cannot manufacture significance.
- Bootstrap p-values are floored at $1/(n+1) = 0.00005$. Two hypotheses sit on that floor, so their true values are unresolved below it.
- **This corrects overlap only.** Survivorship — roughly 46% observability across the EDGAR seam — is untouched, remains the binding limitation, and every row above still carries the survivorship-biased stamp. Chapter 8 is the one that actually hurts.

Free data deletes the evidence against you

This is the finding that generalises beyond tender offers, beyond trading, and it is the one I would keep if I could keep only one.

Four strategies died the same death, and it took me far too long to see it as one thing:

strategy	what was missing	why
insider-buy event study	45.8% of events unpriceable	the company later died
US merger arbitrage	8.6% of completed deals priceable, vs 53.3% of non-completed	completing delists the target
FTSE 250 index changes	33% unpriceable in the best era	the company later died
odd-lot tender arbitrage	profitable deals vanish from ticker sources	the tender delists the company

Free data deletes exactly the observations that would falsify the thesis, and the bias always runs flattering.

This is not bad luck. Event-driven strategies are structurally hostile to free data, because *the event that makes them interesting is the event that removes the entity from the sources a retail researcher can reach*. The company that went bankrupt is the one missing from your insider-buy sample. The merger that completed is the one missing from your arb sample. You are not sampling from the population; you are sampling from the survivors, and the survivors are the ones your thesis is right about.

The screen that follows costs ten minutes and would have saved me weeks:

If this strategy failed on a given observation, would I still see that observation in my data?

Applied to eight mechanisms, it separated them cleanly: every failure was event-driven, and every pass was a property of a currently-existing thing — posted rates, published costs, live spreads.

Demand the coverage-by-outcome table before the result table. If coverage differs by outcome, the headline number is not interpretable — not "less reliable", not interpretable.

The worked example

Merger arbitrage reduces to a single question: can wide-spread deals be selected without their break rate rising proportionally? The threshold is derived, not chosen: **9.8%**.

Measured on 588 deals, the wide bucket showed **6.4%**. That looks like a pass. It looks like a strategy.

But proxy coverage was **15.2% for broken deals against 27.6% for completed** — breaks under-sampled by 1.8x. Correcting for it gives **11.0%**, above the bar.

The uncorrected number would have been a live trade.

Chapter 9

What it earns, and why I could not earn it

What survived requires no forecast

Almost nothing survived, and the survivors share one property: none of them predicts anything.

Cost of ownership. Dividend withholding tax suffered *inside* a fund is deducted before NAV is struck. It appears in no TER, on no factsheet, and in no comparison table. For US equity exposure it routinely exceeds the entire headline fee of a cheap tracker. Ranking funds on TER alone therefore gets the answer wrong, and gets it wrong invisibly — the number you are optimising is not the number you are paying.

Posted rate differentials. GBP overnight-rate ETFs returned roughly 4.2–4.4% against 3.60% on broker cash. Certain, and linear in capital.

A cost saving is deducted whether markets rise or fall, compounds every year it is held, and is knowable today. That asymmetry is why these two survived when every forecasting strategy did not. They do not require me to be right about the future. They require me to have read the right document.

The integrated number

Every component measured, corrected and combined, at £1,500 of capital:

component	£/yr	basis
odd-lot cash tenders	£72	6 tradeable deals over 3 years, Dutch auctions priced at midpoint, positions sized to £1,500
split-off exchange offers	£82–139	~1/yr; mean of 4 = +5.5%, median = +9.29%
idle cash (~9 months)	£40	3.60% AER
total	£195–252/ yr	13.0–16.8%/yr = £16–21/month
all-cash alternative	£54/yr	3.60% = £4.50/month
edge over cash	£141–198/ yr	£12–16/month

Capital is deployed about three months a year and sits in cash the rest of the time, which is why the strategy's own cash line is smaller than the all-cash alternative it is measured against.

Why the odd-lot line is £72 and not £180

Chapter 6 landed on £15/month for odd-lot tenders. This table says £72/year, which is £6/month. Both are correct, and the gap between them is the most important number in the chapter.

Chapter 6 measures the mechanism. Every takeable deal, 99 shares each, priced against the true low of the range. It answers: how much value does odd-lot priority create?

This table measures what £1,500 can reach. And 99 shares is not a small position — at \$40 a share it is roughly £3,100, twice the entire account. Across the sample, a single lot ties up anywhere from £400 to £3,100 for days to weeks. Most qualifying deals simply cannot be funded, and two that overlap cannot both be taken.

The mechanism is worth about £180/year. The account can capture about £72 of it. **The missing £108 is not a market inefficiency, a tax, or a fee. It is the account being too small to show up.**

One honesty note on the comparison: this table prices Dutch auctions at the **midpoint** where chapter 6 uses the conservative **low**. That convention pushes the £72 *up*, not down — so the capital constraint bites even harder than the difference between the two figures suggests.

Thirteen to seventeen per cent a year is not a bad answer. It is a *real* answer, arrived at through pre-registration, FDR correction across every hypothesis ever tested, a sealed holdout, an observability screen, three parser rewrites and a field-by-field audit against live SEC documents. Of everything in this book, that table is the part that survived the most scrutiny.

The broker settled it in one message

On 3 August 2026 I asked Trading 212 support how to submit a tender election. Verbatim:

"I'm sorry, but we do not currently provide access to tender offers or buyback events. Therefore, you won't be able to submit a tender election as you would for a rights issue. Regarding fractional shares, only whole shares can be transferred, so fractional shares cannot be tendered."

Both earning components in that table require submitting a tender election.

line	£/yr as measured	status in the account
odd-lot cash tenders	£72	unexecutable
split-off exchange offers	£82–139	unexecutable
idle cash	£54	unaffected — it is just cash
edge over cash	£141–198	£0

£16–21/month becomes £4.50/month, and the £4.50 is the cash rate, not a strategy.

Every mechanism that survived every statistical control in this book was defeated by a feature that does not exist in an app.

What this does not kill

The mechanism. Only the venue. Nothing measured about odd-lot tenders is invalidated by any of the above:

- odd-lot priority is a real provision, verified on five cached SEC filings that genuinely paid;
- the detection pipeline works — clause detection went from 20% to 100% after the markup-stripping fix in chapter 5;
- the per-deal economics stand.

The blocker is a broker capability, not a market fact. That makes it, oddly, the most fixable problem in the entire project — and the only one whose solution is a decision rather than a discovery. Every other wall here was made of evidence. This one is made of a product roadmap.

Two reasons not to treat it as final: it was a **bot** reply rather than a human agent, which is worth one escalation; and it says "*currently*", which is either boilerplate or a hint. Neither justifies quoting the old number in the meantime. The honest default is that the capability is absent until a human says otherwise.

So the scanner keeps running. It costs nothing, it works, and if a second broker appears it is immediately useful. Its alerts now state the execution requirement explicitly, so a fired alert can never be mistaken for an actionable trade.

The question this leaves

Which broker supports voluntary corporate-action elections for a UK retail account, ideally inside an ISA?

Interactive Brokers is the obvious candidate and is *believed* to have a voluntary-corporate-action election system. That is not verified, and it must be before any account is opened — this book should not end by making the same class of assumption it spent nine chapters dismantling. The ISA question is separate again: a broker may support tenders in a general account and not inside the wrapper.

That is a research question with a concrete answer, and it gates everything the project measured.

The binding constraint was never the code. It is capital, deal flow, and finally a checkbox. More engineering does not move any of the three.

A note on the arithmetic in this chapter. Every figure above is taken from a single integrated measurement dated 3 August 2026, which supersedes the piecemeal numbers produced earlier that day — several of which were wrong, in the ways recorded throughout this

book. Earlier drafts of the project log carry two other totals on slightly different bases: one including a Canadian share-buyback line and counting twelve months of cash rather than nine, and one stating the certain surface at $\sim$ £65/year rather than £54. Those are not corrections to what is above; they are different decompositions of the same underlying deals, and quoting them interchangeably is exactly the error chapter 4 catalogues twenty-one times. One basis, stated once, is the only defensible way to end a book like this.

Chapter 10

What I would tell someone starting this

1. **Ask the observability question before writing the scanner.** It is the cheapest filter available and it dominates every statistical control that comes after.
2. **Compute your multiple-testing bar first.** If you cannot state m , you are not testing hypotheses, you are shopping.
3. **Check every aggregate one level finer before believing it.** I made this error seven times in a single day: counting filings instead of companies, rows instead of episodes, releases instead of deals. Every time, the wrong number was the flattering one. Knowing the rule was not enough — it had to become a tool that refuses to return a rate without the entity count beside it.
4. **A statistic computed at the wrong grain arrives with the authority of real data.** It is more dangerous than a guess, not less.
5. **Verify the execution path before you verify the edge.** Ten minutes with broker support would have reordered eight months of work.

Closing

The pipeline did its job. It killed almost everything, honestly, including several results I had already written up as wins.

The methodology is the deliverable. The P&L is the evidence it was applied.

Nothing described here traded real money. All results are paper or backtest.